

SYSTEM OF CIRCLES

Introduction

A set of circles is said to be a *system of circles* if it contains at least two circles. If two circles touch each other then there exists only one tangent at the point of contact of the circles.

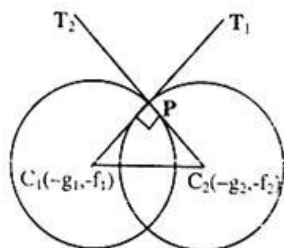


Figure 1: Orthogonal Circle

Two intersecting circles are said to cut each other *orthogonally* if the angle between the circles is a right angle (*S. L. Loney, Article 182, page 160*).

In Figure 1, C_1T_1 is a tangent line to the first circle while C_2T_2 is a tangent line to the second circle and the angle at the point of contact made by the tangents to the circles is 90° . So they are *orthogonal circles*.

Angle of intersection of two circles

The angle of intersection of two circles is the angle between their tangents at a point of intersection.

Let the equations of the circles be

$$x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0 \quad (2)$$

$$x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0 \quad (3)$$

Then $r_1^2 = g_1^2 + f_1^2 - c_1$, $r_2^2 = g_2^2 + f_2^2 - c_2$ and $d^2 = (g_1 - g_2)^2 + (f_1 - f_2)^2$.

$$\begin{aligned} \text{Hence } \cos \theta &= \frac{r_1^2 + r_2^2 - d^2}{2r_1r_2} \\ &= \frac{g_1^2 + f_1^2 - c_1 + g_2^2 + f_2^2 - c_2 - \{(g_1 - g_2)^2 + (f_1 - f_2)^2\}}{2\sqrt{g_1^2 + f_1^2 - c_1} \sqrt{g_2^2 + f_2^2 - c_2}} \end{aligned} \quad (4)$$

Therefore, two circles intersect each other orthogonally if

$$2g_1g_2 + 2f_1f_2 - c_1 - c_2 = 0 \Rightarrow 2g_1g_2 + 2f_1f_2 = c_1 + c_2.$$

Example 1

Find the angle of intersection of the circles $x^2 + y^2 + 4x - 5 = 0$ and $x^2 + y^2 + 4x - 14y + 28 = 0$.

Solution. Let θ be the angle of intersection of the given circles.

$$\begin{aligned}\text{Then we have } \cos \theta &= \frac{2g_1g_2 + 2f_1f_2 - c_1 - c_2}{2\sqrt{g_1^2 + f_1^2 - c_1} \sqrt{g_2^2 + f_2^2 - c_2}} \\ &= \frac{2 \times 2 \times 2 + 2 \times 0 \times (-7) + 5 - 28}{2\sqrt{9}\sqrt{25}} = \frac{-15}{2 \times 15} = -\frac{1}{2}\end{aligned}$$

$$\text{Thus, } \theta = \frac{2\pi}{3}.$$

Example 2

Show that the circles $x^2 + y^2 - 4x - 6y + 4 = 0$ and $x^2 + y^2 - 10x - 14y + 58 = 0$ are orthogonal to each other.

Solution. We know that the circles

$$x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0 \quad (1)$$

$$\text{and} \quad x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0 \quad (2)$$

are orthogonal if $2g_1g_2 + 2f_1f_2 = c_1 + c_2$

Since in the problem we find $g_1 = -2$, $f_1 = -3$, $g_2 = -5$, $f_2 = -7$, compute

$$2g_1g_2 + 2f_1f_2 - c_1 - c_2 = 2(-2)(-5) + 2(-3)(-7) - 4 - 58 = 0$$

Hence the given circles are orthogonal.

(S.L. Loney, Article 190, page 167: Orthogonal Circles)

Example 3

Find the equation of the circle which passes through (1, 1) and cuts orthogonally each of the circles $x^2 + y^2 - 8x - 2y + 16 = 0$ and $x^2 + y^2 - 4x - 4y - 1 = 0$.

$$\text{Solution. Let the required circle be } x^2 + y^2 + 2gx + 2fy + c = 0 \quad (1)$$

Since (1) passes through (1, 1), we have

$$2 + 2g + 2f + c = 0 \Rightarrow 2g + 2f + c + 2 = 0 \quad (2)$$

Also, since (1) cuts orthogonally each of the given circles, we have

$$2g(-4) + 2f(-1) - c - 16 = 0 \Rightarrow -8g - 2f - c - 16 = 0 \quad (3)$$

$$\text{and } 2g(-2) + 2f(-2) - c + 1 = 0 \Rightarrow -4g - 4f - c + 1 = 0 \quad (4)$$

From (2) and (3), we get $-6g - 14 = 0 \Rightarrow g = -\frac{7}{3}$ and from (3) and (4), we get

$$4g - 2f + 17 = 0 \Rightarrow 4\left(-\frac{7}{3}\right) - 2f + 17 = 0 \Rightarrow f = \frac{23}{6}$$

So from (2), we get $c = -5$.

Hence the required equation is $3(x^2 + y^2) - 14x + 23y - 15 = 0$.

Example 4

Find the equation of the circle which cuts orthogonally the circles $x^2 + y^2 + 4x + 7 = 0$, $2x^2 + 2y^2 + 3x + 5y + 9 = 0$ and $x^2 + y^2 + y = 0$.

Solution. Let the equation of the required circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (1)$$

Since it cuts the three given circles orthogonally, we have

$$4g = c + 7, \quad \frac{3}{2}g + \frac{5}{2}f = c + \frac{9}{2}, \quad \text{and } f = c.$$

Solving them, we get $g = 2, f = 1, c = 1$.

Hence the required circle is given by $x^2 + y^2 + 4x + 2y + 1 = 0$.

Example 5

Find the equation of the circles which intersect $x^2 + y^2 - 6y + 1 = 0$ and $x^2 + y^2 - 4y + 1 = 0$ orthogonally and touch the line $3x + 4y + 5 = 0$.

Solution. Given $x^2 + y^2 - 6y + 1 = 0 \quad (1)$

$$\text{and } x^2 + y^2 - 4y + 1 = 0 \quad (2)$$

Let the equation of the required circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad (3)$$

Applying the condition of orthogonality, we get

$$2g \times 0 + 2f(-3) = c + 1 \Rightarrow 6f + c + 1 = 0 \quad (4)$$

$$\text{and } 2g \times 0 + 2f(-2) = c + 1 \Rightarrow 4f + c + 1 = 0 \quad (5)$$

Solving (4) and (5), we get $f = 0$ and $c = -1$.

$$\text{So with } f = 0 \text{ and } c = -1, (3) \text{ becomes } x^2 + y^2 + 2gx - 1 = 0 \quad (6)$$

whose centre is at $(-g, 0)$ and radius is $\sqrt{g^2 + 1}$. If (6) touches the given line, then

$$\frac{3(-g) + 5}{\sqrt{9 + 16}} = \sqrt{g^2 + 1} \Rightarrow -3g + 5 = 5\sqrt{g^2 + 1}$$

$$\Rightarrow 16g^2 + 30g = 0 \Rightarrow g = 0, -\frac{15}{8}.$$

Hence the required circles are

$$x^2 + y^2 = 1 \text{ and } x^2 + y^2 + 2\left(-\frac{15}{8}\right)x - 1 = 0 \Rightarrow 4(x^2 + y^2) - 15x - 4 = 0.$$

Radical Axis

(S. L. Loney, Article 183, page 161)

The radical axis of two circles is the locus of a point which moves so that the lengths of tangents drawn from it to the two circles are equal.

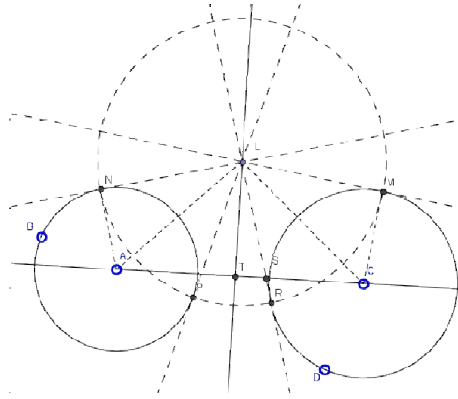


Figure 2: Radical Axis of Two Circles

Radical Axis of two circles

Given $x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$ (2)

$$x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0 \quad (3)$$

Let $P(x_1, y_1)$ be any point on the radical axis from which the lengths of the tangents to (2) and (3) are equal. Therefore, we have

$$\begin{aligned} x_1^2 + y_1^2 + 2g_1x_1 + 2f_1y_1 + c_1 &= x_1^2 + y_1^2 + 2g_2x_1 + 2f_2y_1 + c_2 \\ \Rightarrow 2(g_1 - g_2)x_1 + 2(f_1 - f_2)y_1 + (c_1 - c_2) &= 0 \end{aligned} \quad (5)$$

Hence, the locus of $P(x_1, y_1)$ is

$$2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$$

which is the equation of a straight line.

Corollary

- (i) If $S_1 = 0$ and $S_2 = 0$ be the equation of two circles, then the radical axis of the two circles is $S_1 - S_2 = 0$, i.e. the radical axis passes through the common points, real or imaginary, of the circles $S_1 = 0$ and $S_2 = 0$. (*S. L. Loney, Article 184, 185; page 163*)

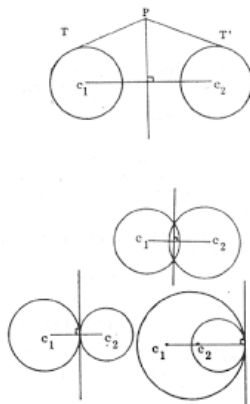


Figure 3: Radical axis passes through the imaginary and real common points

- (ii) The *radical axis* of two intersecting circles is their common chord, real (when the circles meet in real points) or imaginary (when the line passes through the imaginary points of the intersection of the two circles $S_1 = 0$ and $S_2 = 0$).

(*S. L. Loney, Article 184, 185; page 163*)

- (iii) The *radical centre* is the point at which the radical axes of the three circles meet (i.e., the point of concurrence of the radical axes of the three circles).

(*S. L. Loney, Article 186, page 163: The three radical axes of three circles, taken in pairs, meet in a point*)

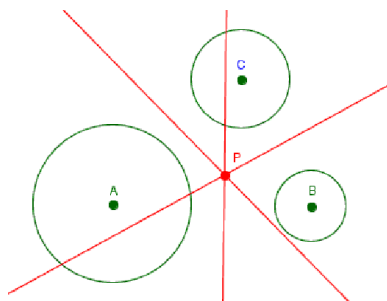


Figure 4: Radical Center

Let the three circles in Figure 4 be called A, B and C and let the radical axis of A and B; the radical axis of B and C and that of A and C meet in a point P, called the radical center.

By definition of the radical axis,

the tangent from P to the circle A = the tangent from P to the circle B and

the tangent from P to the circle B = the tangent from P to the circle C

Hence, the tangent from P to the circle A = the tangent from P to the circle C,

i.e. P is also a point on the radical axis of the circles A and C.

Thus, the radical axes of three circles, taken in pairs, are concurrent.

Properties of radical axis

(i) The radical axis of two circles is a line perpendicular to the line joining the centres of the circles.

(ii) The radical axes of three circles, taken in pairs, are concurrent.

(iii) The radical axis of two circles bisects each of their common tangents (namely, transverse and direct common tangents).

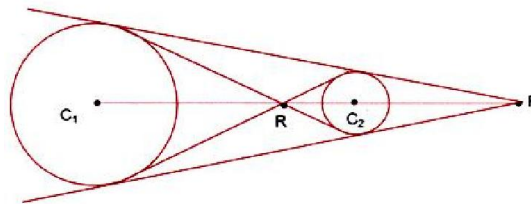


Figure 5: Direct and Transverse Common Tangents

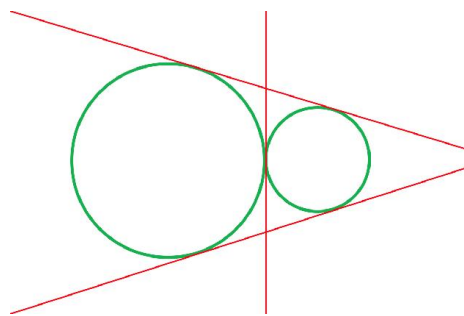


Figure 6: The Radical axis bisects each of their common tangents

(iv) The lengths of the tangents from the radical centre to the three circles are equal.

(v) Radical axis is the locus of the centre of the circle cutting two given circles orthogonally.

Coaxial circles

A system of circles is said to be *coaxial* when the circles of the system have a common radical axis.

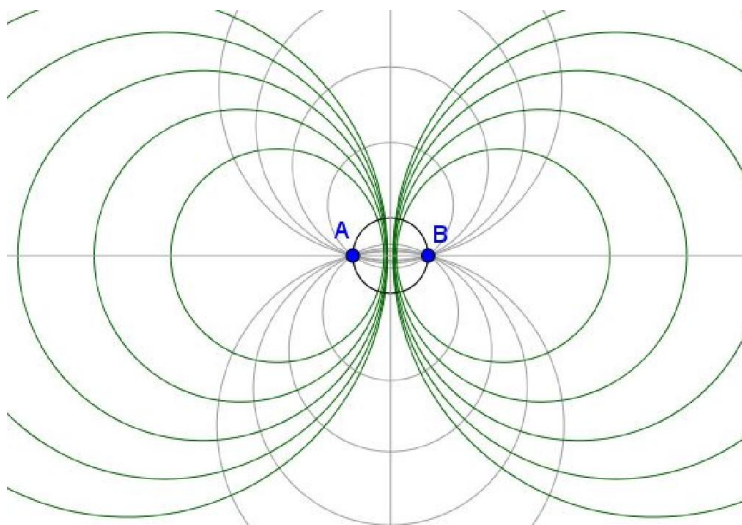


Figure 7: Coaxial circles

The general equation of a system of coaxial circles can be written in the following prescribed forms:

(I) If $S_1 = 0$ and $S_2 = 0$ be the equation of any two circles having $S_1 - S_2 = 0$ as the radical axis, then $S_1 + \lambda S_2 = 0$, where λ is an arbitrary constant.

(S. L. Loney, Article 187, page 164)

(II) If $S_1 = 0$ and $S_2 = 0$ be the equation of any two circles having $S_1 - S_2 = 0$ as the radical axis, then $S + \lambda(S_1 - S_2) = 0$, where λ is an arbitrary constant, and $S = 0$ is any circle of the system.

Example 6

Find the equation of the circle which passes through $(3, -2)$ and coaxial with each of the circles $x^2 + y^2 + 4x - 3y - 7 = 0$ and $x^2 + y^2 + 3x - 2y - 10 = 0$.

Solution. Given $x^2 + y^2 + 4x - 3y - 7 = 0$ (1)

and $x^2 + y^2 + 3x - 2y - 10 = 0$ (2)

The equation of the circle coaxial with (1) and (2) is

$$S_1 + \lambda S_2 = 0 \Rightarrow (x^2 + y^2 + 4x - 3y - 7) + \lambda (x^2 + y^2 + 3x - 2y - 10) = 0 \quad (3)$$

Since it passes through (3, -2), we get

$$\begin{aligned}
 3^2 + (-2)^2 + 4 \times 3 - 3(-2) - 7 + \lambda \{3^2 + (-2)^2 + 3 \times 3 - 2(-2) - 10\} &= 0 \\
 \Rightarrow 24 + 16\lambda &= 0 \\
 \Rightarrow \lambda &= -\frac{3}{2}
 \end{aligned}$$

Hence the required equation is $(x^2 + y^2 + 4x - 3y - 7) + (-\frac{3}{2})(x^2 + y^2 + 3x - 2y - 10) = 0$

$$\begin{aligned}
 \Rightarrow 2(x^2 + y^2 + 4x - 3y - 7) - 3(x^2 + y^2 + 3x - 2y - 10) \\
 \Rightarrow -x^2 - y^2 - x + 16 &= 0 \\
 \Rightarrow x^2 + y^2 + x - 16 &= 0.
 \end{aligned}$$

Example 7

Find the radical centre of the circles

$$x^2 + y^2 + x + 2y + 3 = 0 \quad (1)$$

$$x^2 + y^2 + 2x + 4y + 5 = 0 \quad (2)$$

$$x^2 + y^2 - 7x - 8y - 9 = 0. \quad (3)$$

Solution: The radical axis of the circles (1) and (2) is given by

$$x^2 + y^2 + x + 2y + 3 - (x^2 + y^2 + 2x + 4y + 5) = 0 \Rightarrow x + 2y + 2 = 0 \quad (4)$$

and the radical axis of the circles (1) and (3) is given by

$$x^2 + y^2 + x + 2y + 3 - (x^2 + y^2 - 7x - 8y - 9) = 0 \Rightarrow 4x + 5y + 6 = 0 \quad (5)$$

Solving equations (4) and (5), we get $x = y = -\frac{2}{3}$, which is the desired radical centre of the given circles.

Example 8

Find the equation of the circle whose diameter is the common chord of the circles

$$x^2 + y^2 + 2x + 3y + 1 = 0 \quad (1)$$

and $x^2 + y^2 + 4x + 3y + 2 = 0. \quad (2)$

Solution: The equation of the common chord (i.e. the radical axis) of (1) and (2) is

$$x^2 + y^2 + 2x + 3y + 1 - (x^2 + y^2 + 4x + 3y + 2) = 0 \Rightarrow 2x + 1 = 0 \quad (3)$$

Now any circle through the extremities of this chord is

$$x^2 + y^2 + 2x + 3y + 1 + k(2x + 1) = 0 \Rightarrow x^2 + y^2 + (2 + 2k)x + 3y + (1 + k) = 0 \quad (4)$$

The common chord (3) will be the diameter of (4) if its centre $(-1 - k, -\frac{3}{2})$ lies on (3).

Therefore, using (3) we get $2(-1 - k) + 1 = 0$ giving $k = -\frac{1}{2}$.

Thus, the required equation of the circle is $x^2 + y^2 + x + 3y + \frac{1}{2} = 0$.

Point circles and limiting points

A circle whose radius is zero is called a *point circle*.

A *limiting point* of a system of coaxial circles are the centres of the point circles belonging to the system.

The limiting points of the system of circles represented by the equation

$$x^2 + y^2 + 2\lambda x + c = 0 \Rightarrow (x + \lambda)^2 + y^2 = \lambda^2 - c \quad (6)$$

which represents a system of coaxial circles whose common radical axis is $x = 0$ and whose centres lie on $y = 0$. The centre of (1) is $(-\lambda, 0)$ and radius is $\sqrt{\lambda^2 - c}$. For limiting points, we write $\sqrt{\lambda^2 - c} = 0 \Rightarrow \lambda = \pm\sqrt{c}$.

Therefore, the point circles of the coaxial system are given by

$$(x - \sqrt{c})^2 + y^2 = 0 \quad (7)$$

$$\text{and } (x + \sqrt{c})^2 + y^2 = 0 \quad (8)$$

The centres of (7) and (8) are $(\sqrt{c}, 0)$ and $(-\sqrt{c}, 0)$, respectively, which are the limiting points of (1).

The limiting points are real, coincident or imaginary according as c is greater than, equal, or less than zero.

Properties of limiting points

- (i) The polar of either limiting points with respect to any circle of the system of coaxial circles passes through the other.
- (ii) Every circle through the limiting points of a coaxial system is orthogonal to all circles of the system.

Example 9

Find the limiting points of the system of circles coaxial with the circles $x^2 + y^2 - 6x - 6y + 4 = 0$ and $x^2 + y^2 - 2x - 4y + 3 = 0$.

Solution: Given $x^2 + y^2 - 6x - 6y + 4 = 0 \quad (1)$

and $x^2 + y^2 - 2x - 4y + 3 = 0 \quad (2)$

The equation of the system of circles coaxial with (1) and (2) is

$$S_1 + \lambda S_2 = 0 \Rightarrow (x^2 + y^2 - 6x - 6y + 4) + \lambda (x^2 + y^2 - 2x - 4y + 3) = 0 \quad (3)$$

$$\Rightarrow (1 + \lambda)(x^2 + y^2) - 2(3 + \lambda)x - 2(3 + 2\lambda)y + (4 + 3\lambda) = 0$$

$$\Rightarrow x^2 + y^2 - \frac{2(3 + \lambda)}{1 + \lambda}x - \frac{2(3 + 2\lambda)}{1 + \lambda}y + \frac{4 + 3\lambda}{1 + \lambda} = 0 \quad (4)$$

So, radius of the circle (4) is

$$R = \sqrt{\frac{(3 + \lambda)^2}{(1 + \lambda)^2} - \frac{(3 + 2\lambda)^2}{(1 + \lambda)^2} - \frac{4 + 3\lambda}{1 + \lambda}} = \sqrt{(3 + \lambda)^2 + (3 + 2\lambda)^2 - (4 + 3\lambda)(1 + \lambda)}$$

and centre of (4) is $\left(\frac{3 + \lambda}{1 + \lambda}, \frac{3 + 2\lambda}{1 + \lambda}\right)$.

For limiting points, $R = 0 \Rightarrow \sqrt{(3 + \lambda)^2 + (3 + 2\lambda)^2 - (4 + 3\lambda)(1 + \lambda)} = 0$

$$\Rightarrow (3 + \lambda)^2 + (3 + 2\lambda)^2 - (4 + 3\lambda)(1 + \lambda) = 0$$

$$\begin{aligned}
&\Rightarrow 9 + 6\lambda + \lambda^2 + 9 + 12\lambda + 4\lambda^2 - 4 - 7\lambda - 3\lambda^2 = 0 \\
&\Rightarrow 2\lambda^2 + 11\lambda + 14 = 0 \qquad \Rightarrow 2\lambda^2 + 4\lambda + 7\lambda + 14 = 0 \\
&\Rightarrow 2\lambda(\lambda + 2) + 7(\lambda + 2) = 0 \qquad \Rightarrow (\lambda + 2)(2\lambda + 7) = 0 \\
&\Rightarrow \lambda = -2 \text{ or } \lambda = -\frac{7}{2}
\end{aligned}$$

Therefore, the limiting points are $\left(\frac{3-2}{1-2}, \frac{3-4}{1-2}\right), \left(\frac{3-\frac{7}{2}}{1-\frac{7}{2}}, \frac{3-2 \times \frac{7}{2}}{1-\frac{7}{2}}\right)$ or $(-1, 1), \left(\frac{1}{5}, \frac{8}{5}\right)$.

Exercises

- Find the angle of intersection of the circles $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 = 16$.
- Find the value of k if the circles $x^2 + y^2 + kx + 3y - 5 = 0$ and $x^2 + y^2 + 5x + ky + 7 = 0$ cut orthogonally.
- Find the equation of the circle which cuts orthogonally the circles $x^2 + y^2 + 8x + 10 = 0$, $x^2 + y^2 + 7y + 7 = 0$ and $x^2 + y^2 + 4x + 6y + 16 = 0$.
- Prove that the two circles each of which passes through $(0, k)$ and $(0, -k)$ and touch the line $y = mx + b$ will cut orthogonally if $b^2 = k^2(2 + m^2)$.
- Find the equation of the circle which passes through the origin and belongs to the coaxial system whose limiting points are $(1, 2)$ and $(3, 4)$.
- Find the radical axis of the coaxial system of circles whose limiting points are $(-1, 2)$ and $(2, 3)$.
- Prove that the limiting points of the system $x^2 + y^2 + 2gx + c + \lambda(x^2 + y^2 + 2fy + k) = 0$ subtend a right angle at the origin if $\frac{c}{g^2} + \frac{k}{f^2} = 2$.